The SUETRI-A-Team

What Should We Count? Token Accounting for the Carbon Footprint of Agentic AI

Ge Zhang, Jingru Cheng, Zitong Huang, Zhibo Dai, and Anthony R. Kovscek
Department of Energy Science & Engineering, Stanford University

Software Engineering FTE to Token Consumption

Within each stage: light · 0.30 FTE/day | medium · 0.50 FTE/day | dark · 0.70 FTE/day — Thin paths: sampled Monte Carlo days



Modeling | conditional high-load case (one engineer-day)

System boundary
Inference + Facility Electricity

1 Work to sessions
20 agent sessions/day
40 equal-effort units × q
N ~ Binomial(40, q); attempted FTE/day = N/40

2 Sessions to tokens
35.1M tokens/day
1.4M median total tokens/session
Session-token spread: lognormal σ = 0.7

3 Tokens to energy
5.94 H100 GPU-h/day
1,642 total tokens/s at 326 W board
Facility energy: 3.25 kWh/day (host × PUE)

4 Energy to emissions
U.S. HDC mix
0.325 kg CO_2e/kWh
IPCC source LCA × U.S. supply shares
Electricity LCA; EPA operational kept separate

Final outputs
1.05 kg CO_2e/day
3.25 kWh/day | 35.1M tokens/day
P10–P90: 0.72–1.46 kg/day
Attributional electricity LCA

How to read
40 candidate work units/day
- q: attempted agent share
- P10–P90: simulated day range
- LCA: electricity life cycle

Conditional high-load scenario

1 Work
40 UNITS/DAY · q = 0.50
0.50
attempted FTE/day
P10–P90 AT q = 0.50
0.40–0.60
N / 40; allocation assumption

2 Agent sessions
40 UNITS/DAY · q = 0.50
20
sessions/day
P10–P90 AT q = 0.50
16–24
N ~ Binomial(40, q)

3 Tokens
40 UNITS/DAY · q = 0.50
35.1
million tokens/day
P10–P90 AT q = 0.50
25.3–46.9
1.4M median per session

4 GPU time
40 UNITS/DAY · q = 0.50
5.94
H100 GPU-hours/day
P10–P90 AT q = 0.50
4.28–7.93
tokens / 1,642 per second

5 Electricity
40 UNITS/DAY · q = 0.50
3.25
facility kWh/day
P10–P90 AT q = 0.50
2.29–4.44
326 W board × host × PUE

6 Grid factor
40 UNITS/DAY · q = 0.50
0.325
kg CO2e/kWh (LCA)
P10–P90 AT q = 0.50
0.286–0.363
U.S. mix × IPCC factors

7 Emissions
40 UNITS/DAY · q = 0.50
1.05
kg CO2e/day (LCA)
P10–P90 AT q = 0.50
0.72–1.46
electricity × grid factor

Example: engineer-day and illustrative global scale-up

40 units/day → 20 sessions → 35.1M tokens → 5.94 H100 GPU-h → 3.25 kWh → 1.05 kg CO_2e/day

Scale-up illustration: 1.05 kg CO_2e/day × 20M professional developers × 365 days/year = 7.665 Mt CO_2e/year.

Assumes daily agent use by every developer; not a forecast or measured net FTE replacement. Day-level LCA P10–P90: 0.72–1.46 kg.

Sources

Sources: Standard Compute; local H100 tests; Guidi et al. (2026); EIA (2024); IPCC AR5; EPA eGRID; JetBrains (2025).